



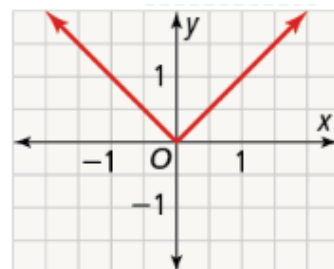
Lesson 1.2: Transformations of Functions

Math Objectives	1. Graph a transformed function by identifying the effect on the graph of replacing $f(x)$ by $f(x) + k$, $kf(x)$, $f(kx)$, and $f(x + k)$ for specific values of k. 2. Write an equation of a transformed function. 3. Relate the domain of a function to its graph and the real-world situation it describes. HSF.BF.B.3, HSF.IF.B.5
Language Objective	Students will explain transformations of functions, provided pre-taught vocabulary, the three-reads strategy, and structured student-talk time.
Essential Question	How do values a , b , h , and k in $a \cdot f[b(x - h)] + k$ transform the graph of a parent function, $f(x)$?

EXPLORE & REASON

The graph of the function $f(x) = |x|$ is shown.

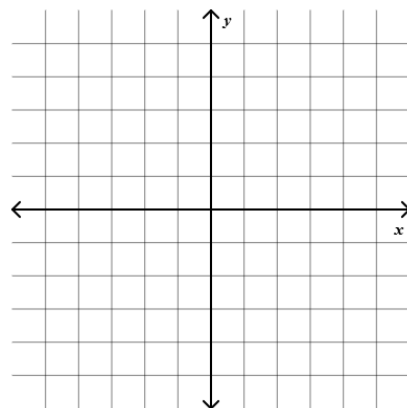
- A. Graph the function $g(x) = |x + c|$ several times with different values of c (any value of -5 to 5).



- B. **Look for Relationships** Predict what will happen to the graph if c is a number greater than 100. What if c is a number between 0 and $\frac{1}{2}$?

EXAMPLE 1 Translate a Function

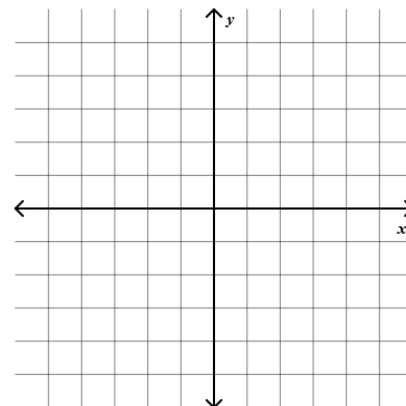
- A. Graph the function $f(x) = x^2$ for the domain $[-2, 2]$.
The graph of g is the graph of f after a translation of 3 units down.
How are the functions, domains, and ranges of f and g related?



B. Graph the function $f(x) = x^2$ for the domain $[-2, 2]$.

The graph of g is the graph of f after a translation 3 units to the right.

How are the functions, domains, and ranges of f and g related?



Try It!

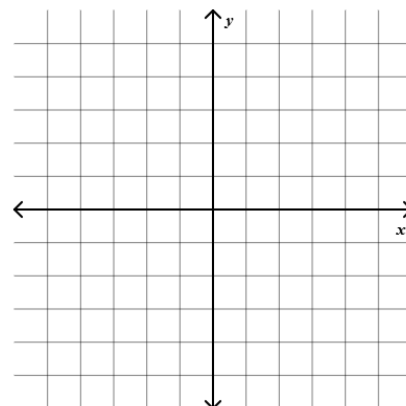
1.

a. Compare the intercepts of f to the intercepts of g in part (a).

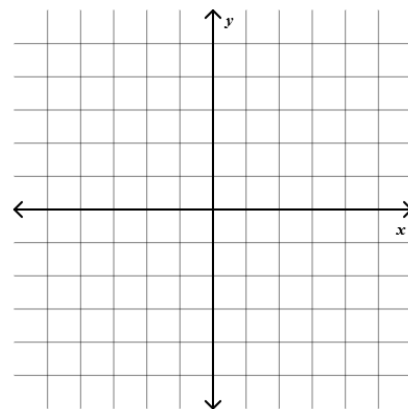
b. Compare the intercepts of f to the intercepts of g in part (b).

EXAMPLE 2 Reflect a Function Across the x - or y -Axis

A. Graph $f(x) = 2x - 6$ for the domain $[0, \infty)$ and the function g , whose graph is the reflection of the graph f across the x -axis. How are their functions related?



- B. Graph $f(x) = 2x - 6$ for the domain $[0, \infty)$ and the function g , whose graph is the reflection of the graph f across the y -axis. How are their functions related?**

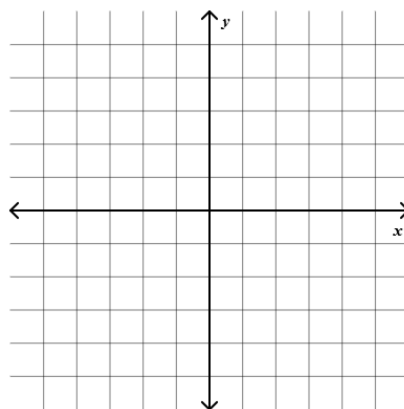
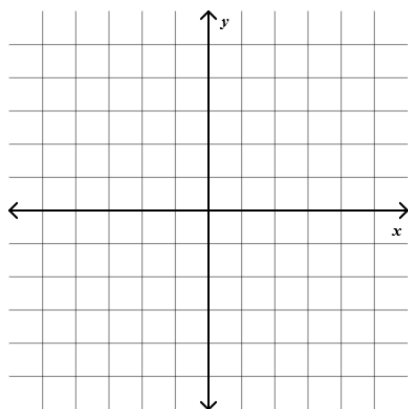


Try It!

- 2.** What is the function for the reflected graph? Check by graphing.

- a.** The graph of $f(x) = x^2 - 2$ reflected across the x -axis.

- b.** The graph of $f(x) = x^2 - 2$ reflected across the y -axis.



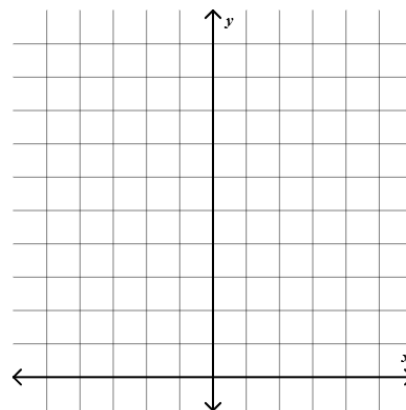
EXAMPLE 3 Understand Stretches and Compressions

A. Graph $f(x) = |x|$ with domain $[-4, 4]$ and $g(x) = 2 \cdot f(x)$.

How are the graphs, domains, and ranges related?

Use a table to find points on the graph(s).

x	$f(x) = x $	$g(x) = 2 \cdot f(x)$	$(x, g(x))$



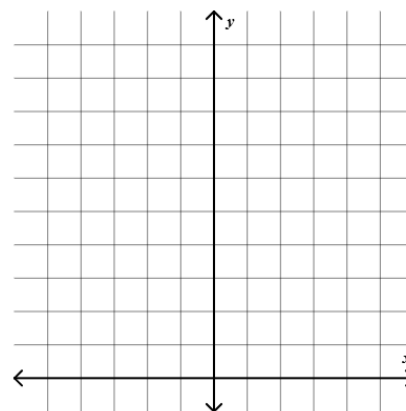
A transformation that increases the distance between the points of a graph and a given line, usually the x -axis, by the same factor, is called a **stretch**.

B. Graph $f(x) = |x|$ with domain $[-4, 4]$ and $h(x) = f(2x)$.

How are the graphs, domains, and ranges related?

Use a table to find points on the graph(s).

x	$f(x) = x $	$h = f(2x)$	$(x, h(x))$

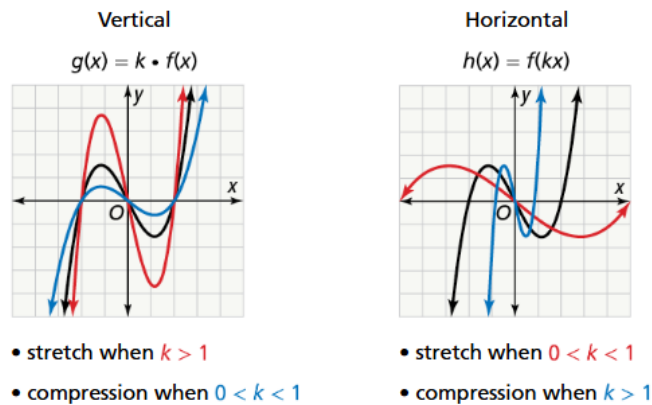


A transformation that decreases the distance between the points of a graph and a given line, usually the y -axis, by the same factor, is called a **compression**.

Try It!

3. The domain of $j(x)$ is $[a, b]$. Show that $j(x) = f(\frac{1}{2}x)$ is a horizontal stretch of the graph of f . What is the domain of f ?

CONCEPT Stretches and Compressions

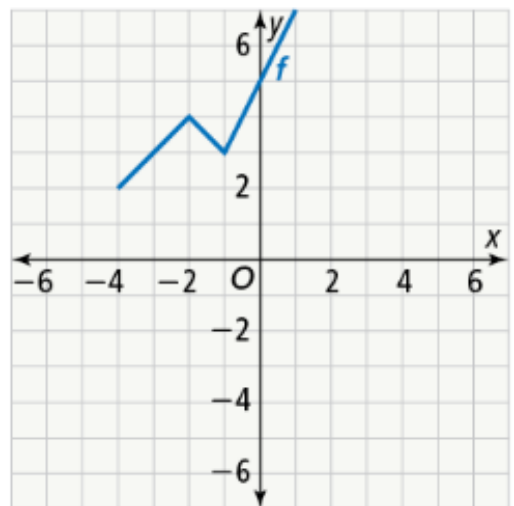


EXAMPLE 4 Graph a Combination of Transformations

Using the graph $y = f(x)$, how can you graph a combination of transformations?

A. Graph $y = -f(x + 2)$.

B. Graph $y = \frac{1}{2}f(x) + 3$.

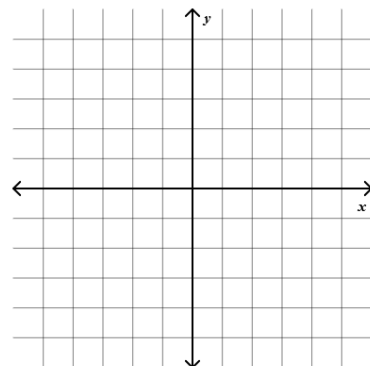
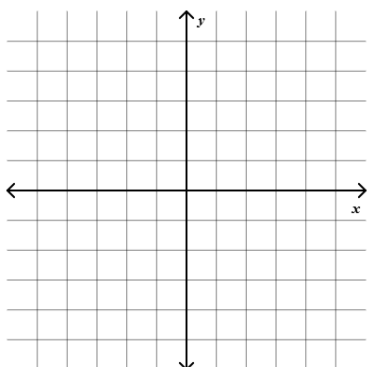


Try It!

4. Using the graph of f above, graph each equation.

a. $y = f(2x) - 4$

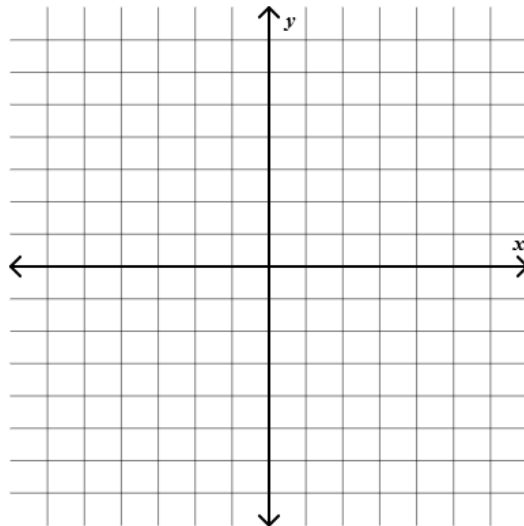
b. $y = f(2x - 3) - 2$



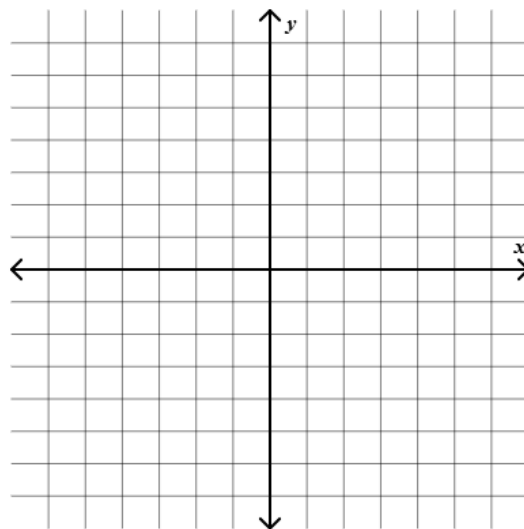
EXAMPLE 5 Identify Transformations From an Equation

What transformations of $f(x) = x^2$ result in the graph of the function g ?

A. $g(x) = -\left(\frac{1}{3}x\right)^2$



B. $g(x) = (x - 4)^2 + 5$

**Try It!**

5. (Collected and graded.) What transformations of the graph of $f(x) = |x|$ are applied to graph the function g ?

a. $g(x) = \frac{1}{2}|x + 3|$

b. $g(x) = -|x| + 2$

CONCEPT SUMMARY Transformations of Functions

For a function $f(x)$, the graph of $f(x) = a \cdot f[b(x - h)] + k$ represents a transformation of the graph of that function by translation, reflection, or stretching.

WORDS	EQUATIONS	GRAPHS
Horizontal translation of f right 2 units (altering h)	$g(x) = f(x - 2)$ $f(x) = x^2 + x$ $g(x) = (x - 2)^2 + (x - 2) = x^2 - 3x + 2$	
Vertical translation of f up 3 units (altering k)	$h(x) = f(x) + 3$ $f(x) = x^2 + x$ $h(x) = x^2 + x + 3$	
Reflection of f across the x-axis (altering a)	$g(x) = -f(x)$ $f(x) = x^2 + x$ $-f(x) = -(x^2 + x) = -x^2 - x$	
Reflection of f across the y-axis (altering b)	$h(x) = f(-x)$ $f(x) = x^2 + x$ $f(-x) = (-x)^2 + (-x) = x^2 - x$	
Horizontal stretch of f by the factor $\frac{1}{2}$ (altering b)	$g(x) = f(\frac{1}{2}x)$ $f(x) = x^2 + x$ $f(\frac{1}{2}x) = (\frac{1}{2}x)^2 + \frac{1}{2}x = \frac{1}{4}x^2 + \frac{1}{2}x$	
Vertical stretch of f by the factor 2 (altering a)	$h(x) = 2f(x)$ $f(x) = x^2 + x$ $2f(x) = 2(x^2 + x) = 2x^2 + 2x$	

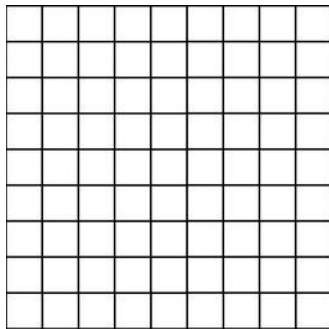
Do You UNDERSTAND?

- Essential Question** How do the values a , b , h , and k in $a \cdot f[b(x - h)] + k$ transform the graph of a parent function, $f(x)$?
- Reason** Do k and h affect the input or output for $g(x) = f(x) + k$ and $g(x) = f(x - h)$? Explain.

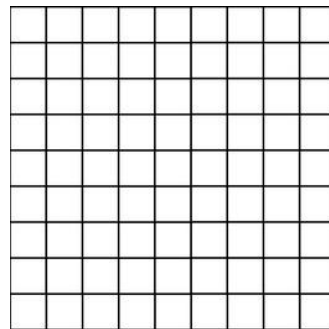
Do You KNOW HOW?

Graph each function and its parent function.

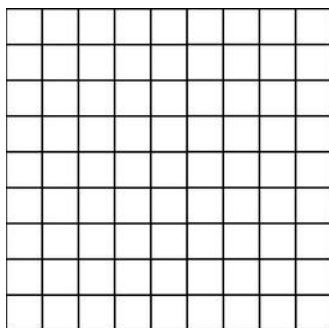
3. $g(x) = |x| - 1$



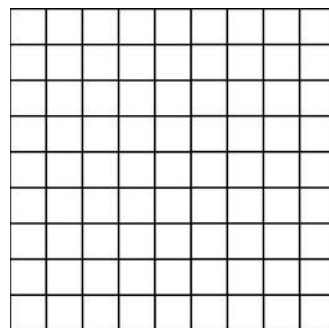
4. $g(x) = (x - 3)^2$



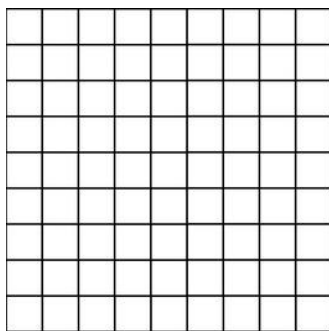
5. $g(x) = -|x|$



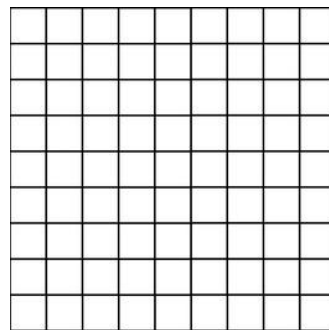
6. $g(x) = -x$



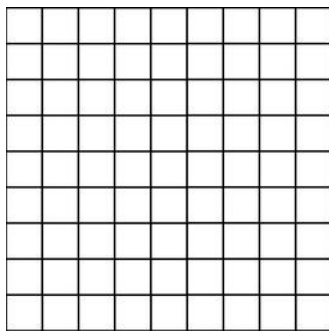
7. $g(x) = x^2 - 2$



8. $g(x) = \frac{1}{2}|x|$



9. $g(x) = 4x$



10. $g(x) = |5x|$

